

COURSE: CIT 316

The Container Revolution: Architecting Local CI/CD Foundations

01. Introduction

In modern software engineering, the phrase *"It works on my machine"* is a relic of the past. DevOps engineers utilize **Virtualization** and **Containerization** to ensure parity between a developer's laptop and a production server. This guide details the steps to prepare a local workstation for professional-grade CI/CD workflows using Docker and Kubernetes.

02. Environment Prerequisites

Before installing software, the hardware must be configured to allow a "Hypervisor" to run.

Hardware Virtualization: Ensure *Intel VT-x* or *AMD-V* is enabled in your BIOS/UEFI. Without this, Docker and Minikube cannot create the isolated environments required for containers.

Verification Command:

```
systeminfo | findstr "Hyper-V"
```

Look for: A hypervisor has been detected.

03. Step-by-Step Setup

1 Install Docker Desktop

Docker is the engine that builds and runs containers. Download and install for Windows/Mac/Linux.

```
docker --version
docker run hello-world
```

2 Initialize Minikube (Local Kubernetes)

Minikube creates a single-node Kubernetes cluster on your machine. It is the gold standard for learning orchestration without cloud costs.

```
minikube start --driver=docker
kubectl get nodes
```

04. Lab 1: Packaging the Application

Transition from a consumer of containers to a creator. We use a **Dockerfile** to define our environment as code.

The Dockerfile (The Infrastructure Recipe)

```
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

Build & Run Commands

```
# Building the image
docker build -t my-first-web-app .

# Running the containerized app
docker run -d -p 8080:80 --name web-server my-first-web-app
```

05. Supplementary Q&A for the Tech-Savvy

Question	The Engineering Perspective
What is the Docker Daemon?	It is the background service (dockerd) that manages images, containers, and networks. The CLI talks to this daemon via an API.
Why use Nginx:Alpine?	Alpine is a security-focused, tiny Linux distribution (approx. 5MB). In DevOps, smaller images mean faster deployments and a smaller attack surface.
What is a Pod vs. a Container?	A container is the runtime. A Pod is a Kubernetes abstraction that can hold one or more containers, sharing a single network IP.